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Process-Based Authorship Assessment Across Human and LLM-Generated Writing: The Content Restoration Authorship Familiarity Test (CRAFT)

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Abstract:

This study examines whether authorship can be distinguished through process-based evidence of engagement with a text, rather than through probabilistic classification of textual features. The widespread adoption of large language model (LLM)-assisted writing challenges the use of student texts as evidence of learning, as existing detection approaches rely on surface-level features and show limited reliability in educational contexts. Whereas prior research has focused on textual features alone, educational contexts allow direct engagement with learners. In such contexts, human authorship entails access to the decisions that produced a text, whereas LLM text generation does not. To operationalize these characteristics of the writing process, the Content Restoration Authorship Familiarity Test (CRAFT) was developed to assess authorship through writers' ability to recall and restore elements of word choice and text structure. Using a within-subjects design, 60 undergraduate English education majors completed CRAFT tasks based on both self-authored handwritten texts and LLM-generated texts on the same topics. Results showed a clear separation between conditions (Cohen's d = 5.34), indicating strong discriminative capacity between the two authorship conditions. Lexical restoration produced larger effects than structural restoration, while their combination yielded substantial separation. These findings suggest that authorship can be operationalized as access to cognitively encoded compositional decisions rather than inferred from textual features, grounding authorship verification in cognitive models of writing and evidence-centered approaches to assessment. The study offers an interpretable alternative to automated detection of LLM-generated text and supports process-based assessment practices that enable instructors to evaluate writers' engagement with planning, drafting, and revision.

Keywords:
authorship verification, large language models (LLMs), writing assessment, process-based assessment, cognitive engagement, academic integrity

1. Introduction

1.1 Writing and assessment in the era of large language models (LLMs)

Text-based compositions are widely used as evidence of learning and as central components of formative and summative assessment across academic disciplines, where student writing is treated as evidence of disciplinary understanding, reasoning, and knowledge construction (Adams et al., 2010; Bazerman, 2016; Lodge et al., 2023). The rapid normalization of LLM-assisted writing has complicated this role, as students now have access to tools capable of generating fluent, contextually appropriate text at scale and with minimal effort, fundamentally altering how written products are produced in educational contexts (Deng et al., 2025; Fernandes et al., 2026; Kasneci et al., 2023; Skulmowski, 2024). While such tools offer clear pedagogical benefits for supporting writing development, particularly for additional-language writers and students engaging in complex academic genres (Barrot, 2023; Deng et al., 2025; Kasneci et al., 2023; Shabir, 2025), they simultaneously challenge the integrity of writing-based assessment by obscuring the relationship between submitted texts and students' own cognitive engagement and authorship (Chen & Pi, 2026; Draxler et al., 2024; Skulmowski, 2024). In response, institutions and instructors have adopted a range of strategies, including prohibitions on LLM use, reliance on automated detection tools, and honor-based disclosure policies (Janinovic et al., 2024; Jin et al., 2025; McDonald et al., 2025; Wang et al., 2024). However, these approaches are difficult to enforce, often yield probabilistic or opaque judgments, and provide limited pedagogical or diagnostic value for instructional decision-making (Chen & Pi, 2026; Janinovic et al., 2024; Jin et al., 2025; McDonald et al., 2025; Wang et al., 2024). Collectively, these tensions underscore the need for assessment approaches that go beyond enforcement or surveillance, and that support principled distinctions between genuine student authorship and surrogate text production (Lodge et al., 2023).

1.2 Existing approaches to LLM-generated text detection

Existing approaches to detecting LLM-generated text primarily conceptualize detection as a classification task, producing probabilistic judgments about whether a text is human- or LLM-generated rather than definitive evidence of authorship (Fraser et al., 2025; Kehkashan et al., 2025; Pudasaini et al., 2025; Wu et al., 2025; Yang et al., 2025). Across classifier-based, stylometric, and fine-tuned transformer detection approaches, detection performance has been shown to depend heavily on contextual factors such as the specific LLM used for generation, decoding strategy, domain, text length, language, and degree of human involvement. Consequently, the effectiveness of these approaches is undermined by limited robustness and reduced generalizability beyond controlled evaluation settings (Fraser et al., 2025; Kehkashan et al., 2025; Wu et al., 2025). Empirical evaluations of widely used detection tools further report inconsistent classifications and non-trivial false-positive rates on human-written texts, raising concerns about their reliability for academic integrity decisions (Elkhatat et al., 2023; Ghiurău & Popescu, 2025). Studies conducted in educational contexts additionally demonstrate that both automated detectors and human raters struggle to reliably distinguish LLM-generated texts from student-written texts, even under experimental conditions, with judgments often characterized by overconfidence and variability (Barrot & Aranda, 2025; Chen & Pi, 2026; Fleckenstein et al., 2024; Nguyen & Barrot, 2024). As a result, current LLM-generated text detection approaches face persistent challenges related to reliability, generalizability, and interpretability when applied to real-world educational assessment contexts (Fraser et al., 2025; Kehkashan et al., 2025; Liu et al., 2025; Wu et al., 2025; Yang et al., 2025).

1.3 Authorship as cognitive engagement and process-based evidence

Rather than treating authorship as a property of the text itself, this study conceptualizes human authorship (henceforth authorship) as a cognitive relationship between a writer and their written text. Writing is a cognitively demanding activity that relies on working memory to coordinate planning, linguistic encoding, transcription, and revision, drawing on both short-term processing and long-term knowledge representations (Deane et al., 2008; Hayes, 1996; Kellogg et al., 2013; Li, 2023). During composition, writers actively maintain and manipulate content, lexical choices, and structural plans, particularly during higher-level organization and revision (Johnson, 2020; Li, 2023; Lu, 2015; Révész et al., 2019). These processes are shaped by encoding and externalization, as text composition constitutes a process that generates memory traces of prior compositional decisions (Benton et al., 1993; Hayes, 1996; Kellogg et al., 2013). Revision requires writers to review and re-evaluate text across both surface-level form and meaning-level content, drawing on working memory and higher-level planning processes (Adams et al., 2010; Galbraith & Torrance, 2004; Van Waes et al., 2014). Accordingly, composition and revision involve decisions about what was written, why, and how it was structured that are encoded during the writing process (Hayes, 1996; Kellogg et al., 2013; Révész et al., 2017; Van Waes et al., 2014), which can later be accessed and evaluated.

Framed in this way, authorship verification shifts from surface textual features to the underlying cognitive engagement, aligning with construct-valid approaches to assessment that prioritize explanatory adequacy and interpretability (Fulcher, 2024; Mislevy et al., 2006; McNamara, 2001). Given the limitations of probabilistic detection systems (Fraser et al., 2025; Kehkashan et al., 2025; Liu et al., 2025; Wu et al., 2025; Yang et al., 2025), process-based evidence, such as the ability to recall and restore elements of a text, offers a more compelling basis for authorship verification. Given recent concerns about over-reliance on automated evaluation in language assessment in educational contexts (Fulcher, 2024), scholars emphasize eliciting evidence of learners' engagement with planning, decision-making, and revision (Deane et al., 2008; Deane, 2011). From this perspective, authorship can be evidenced through demonstrable access to decisions that originally shaped a text. The present study operationalizes this principle by treating the ability to recall and restore word choice and text structure as observable indicators of human participation in the writing process, offering a process-oriented alternative to probabilistic inference grounded in textual features alone.

1.4 Purpose of the present study

This study investigates whether authorship can be distinguished through process-based evidence of engagement with a text. Specifically, this study evaluates the extent to which the Content Restoration Authorship Familiarity Test (CRAFT) differentiates between human-authored and LLM-generated texts by assessing writers' ability to recall and restore elements of word choice and text structure. To assess robustness across conditions, CRAFT performance is compared following self-authored handwritten texts and LLM-generated texts on the same topic, allowing evaluation of both its discriminative capacity and sensitivity to text condition. Conceptually, this study reframes LLM-generated text detection as a problem of authorship verification grounded in cognitive engagement with the writing process. Practically, this study introduces an interpretable, test-based alternative to automated classifiers by operationalizing authorship as observable access to prior compositional decisions. Pedagogically, this study aligns authorship verification with core principles of writing instruction by foregrounding planning, decision-making, and revision, positioning assessment as an extension of learning-oriented practice rather than mere surveillance. Process-oriented authorship verification may offer a more defensible and educationally meaningful response to LLM-assisted writing.

2. Literature Review

2.1. Automated approaches to authorship analysis

Currently, automated detection of LLM-generated text consists of four main approaches: likelihood-based methods that exploit next-token probability distributions, stylometric feature modeling, supervised neural network-based classification, and generation-time watermarking. Each of these approaches faces persistent challenges in terms of their robustness across contexts (Fraser et al., 2025; Kehkashan et al., 2025; Pudasaini et al., 2025; Wu et al., 2025; Yang et al., 2025). A prominent method exploits statistical regularities associated with next-token generation, using likelihood-, rank-, and perplexity-based measures to characterize differences in how language models and human writers allocate probability across possible next-word choices (Elek et al., 2025; Gritsai et al., 2024). A closely related theme is stylometric/linguistic feature detection, which relies on distributions of lexical, syntactic, or discourse cues and then feeds those features into conventional classifiers, potentially offering greater interpretability or different robustness tradeoffs than purely end-to-end neural approaches that learn representations directly from raw text without explicit feature engineering (Al-Shaibani & Ahmed, 2026; Przystalski et al., 2025; Zaitsu et al., 2025). In parallel, many detection systems are neural network-based classifiers that fine-tune pretrained transformer language models on labeled human versus machine text to learn discriminative representations without an explicit hand-engineered feature stage (Ahmed et al., 2025; Alhijawi et al., 2025; Oghaz et al., 2025). Another major theme is watermarking, which shifts detection upstream by embedding statistically detectable signals at generation time, so detection becomes verification of a designed pattern rather than inference from surface cues (Dathathri et al., 2024; Fraser et al., 2025; Wu et al., 2025; Yang et al., 2025).

2.1.1 Limitations of automated approaches to authorship analysis

Across detection approaches, researchers document a wide range of concrete attack vectors and structural failure modes that undermine detection robustness in realistic settings (Fraser et al., 2025; Kehkashan et al., 2025; Wu et al., 2025; Yang et al., 2025). Text rewriting and paraphrasing attacks feature prominently in this literature, including single-pass and iterative paraphrasing. These attacks alter lexical and syntactic distributions while largely preserving meaning, substantially degrading detection performance across supervised, zero-shot, and watermark-based systems. Related lexical substitution attacks, such as contextual synonym replacement and word swapping, can be implemented through random perturbation or optimized search, but have been shown to evade detectors even when only a small proportion of words are modified (Fraser et al., 2025; Kehkashan et al., 2025). Detectors are further vulnerable to character- and surface-level perturbations, including misspellings, character substitutions, grammatical noise, shuffling, cropping, and deletion, which disrupt statistical signatures without materially affecting readability (Kehkashan et al., 2025; Wu et al., 2025). Several studies highlight tool-mediated paraphrasing using readily available rewriting systems as an accessible and effective evasion strategy (Kehkashan et al., 2025).

Beyond text-level manipulation, prompt-based attacks allow users to evade detection by generating alternative outputs for the same task through prompt variation, bypassing the need for post hoc rewriting (Fraser et al., 2025; Wu et al., 2025). Detection systems are also susceptible to detector-aware optimization, where feedback beyond a binary label (e.g., confidence scores) enables iterative refinement of outputs to minimize detectability (Fraser et al., 2025). Watermarking approaches introduce additional vulnerabilities, including removal through paraphrasing or canonicalization, degradation via translation or polishing, and spoofing attacks in which watermarks are intentionally forged or injected into human-written text (Dathathri et al., 2024; Fraser et al., 2025; Wu et al., 2025). Structural limitations further constrain detection reliability: short texts provide an insufficient signal, reducing accuracy in settings such as short-answer assessment (Yang et al., 2025; Kehkashan et al., 2025), while human-LLM hybrid texts, post-edited outputs, and interleaved authorship violate binary classification assumptions and consistently degrade performance (Fraser et al., 2025; Kehkashan et al., 2025). Furthermore, empirical studies report systematic false positives and bias, raising concerns about fairness and validity in high-stakes educational applications (Fraser et al., 2025; Kehkashan et al., 2025; Wu et al., 2025; Yang et al., 2025). Finally, the literature consistently highlights an "arms-race" dynamic: as detection improves, users (benign or malicious) can adapt prompts, decoding, or post-editing to evade detectors, increasing the importance of threat modeling, realistic benchmarks, and transparent reporting of failure modes (Fraser et al., 2025; Wu et al., 2025; Yang et al., 2025). "In short, there is no clear 'winner' among the available tools" (Fraser et al., 2025, p. 2252).

2.2 Cognition and the writing process

2.2.1 Planning in the independent writing process

Writing planning is a central, cognitively demanding component of composition rather than a peripheral or purely preparatory activity. In their foundational model, Hayes and Flower (1980) conceptualize planning as a goal-directed process involving idea generation, goal setting, and content organization, emphasizing its recursive integration with drafting and revision. Hayes (1996) situates planning within a resource-limited, working-memory framework, showing that its depth and persistence are sensitive to transcription demands, motivation, and cognitive load. Similarly, Kellogg (1987, 1988, 1996) demonstrates that planning competes with drafting and reviewing for limited attentional resources, with strategies such as outlining externalizing structure to reduce later processing demands. Across these models, planning decisions are intentional, strategic, and cognitively encoded, making them potentially retrievable and reportable. Later work (Kellogg, 2001, 2007) further shows that skilled writers maintain global plans and rhetorical schemas through structured retrieval from long-term memory, supporting sentence-level formulation. Empirical studies using keystroke logging, eye-tracking, and stimulated recall provide converging evidence that planning leaves observable temporal and behavioral traces, such as boundary-level pausing and goal-directed attention shifts (Barkaoui, 2019; Révész et al., 2017; Tian et al., 2024). In second language writing, planning plays a critical regulatory role, mediating the effects of working-memory constraints on coherence, argument structure, and revision (Johnson, 2020; Li, 2023; Tabari & Golparvar, 2025). From an assessment perspective, planning constitutes a reflective, strategic mode of cognition that must be explicitly elicited to capture valid writing proficiency, as failures often stem from weak goal formation rather than flawed linguistic encoding (Deane et al., 2008). Accordingly, planning is a memory-encoding, decision-rich process that shapes drafting and revision, and serves as a core locus of cognitive authorship rather than a detachable pre-writing skill.

2.2.2 Planning in the LLM-assisted writing process

LLM-assisted writing reconfigures planning by shifting from internally generated idea formation and goal structuring toward externally mediated coordination and prompt-mediated interaction (Guo et al., 2025; Liu et al., 2024). Rather than engaging in sustained conceptual planning, writers often plan how to interact with the LLM (what to ask, how to frame prompts, and how to sequence outputs), thereby externalizing key planning functions to the interface (Draxler et al., 2024; Yao et al., 2025). This shift is commonly described as distributed cognition, where planning becomes dialogic and reactive, emerging through iterative exchanges with the LLM, instead of the writer's own stable internal representations of content and rhetorical goals (Frenkenberg & Hochman, 2025). Process-tracing studies indicate that while LLMs support rapid idea generation and reduce perceived difficulty, they can reduce depth of content planning, particularly when writers adopt outputs wholesale rather than critically evaluating them (Liu et al., 2024; Zhang et al., 2026). Proficiency differences further moderate this effect: higher-proficiency writers tend to maintain internal goal control and resist delegating planning, whereas lower-proficiency writers are more likely to substitute LLM output for independent planning, gaining fluency without corresponding gains in conceptual engagement (Zhang et al., 2026; Zhao, 2025). At the same time, planning is not eliminated but transformed. When pedagogically scaffolded, learners can sustain metacognitive engagement by articulating goals, refining prompts, and critically interpreting LLM responses (Lee, 2024; Luo et al., 2025). Across studies, however, a consistent tension emerges between efficiency and cognitive depth: while LLMs can extend working memory and accelerate exploration, unstructured reliance may compress or bypass planning processes that encode intentions, support revision, and ground authorship in the writer's cognitive activity (Guo et al., 2025; Skulmowski, 2024).

2.2.3 Drafting in the independent writing process

Cognitive models of writing characterize drafting (variously termed translation, transcription, encoding, or composition) as a resource-intensive process in which conceptual representations are transformed into linguistic form under working-memory constraints (Hayes & Flower, 1980; Kellogg, 1996). Early models emphasized that drafting is not mechanical transcription but a goal-directed activity competing with planning and revision for limited cognitive resources (Hayes & Flower, 1980), later formalized in working-memory frameworks identifying sentence formulation and transcription as primary bottlenecks (Kellogg, 1996, 2013). Empirical studies using dual-task paradigms, keystroke logging, and interference methods show that drafting relies heavily on phonological and executive working memory, with concurrent planning or reviewing degrading fluency and syntactic accuracy (Kellogg et al., 2001; Kellogg et al., 2013; Li, 2023). More recent writing research indicates that drafting efficiency, indexed by burst length, within-burst fluency, and reduced mid-clause pausing, predicts text quality more strongly than pause or revision frequency alone (Révész et al., 2019; Vandermeulen et al., 2024). This effect is especially pronounced in second language writing, where increased encoding demands divert resources from higher-order concerns, causing drafting difficulties to resemble idea-generation problems (Barkaoui, 2019; Li, 2023). Drafting also remains highly recursive, with writers alternating between formulation and micro-planning, particularly in cognitively demanding genres (Tian et al., 2024). Neurocognitive evidence further shows that transcription modality matters: handwriting and typing engage distinct motor and sensory systems that differentially affect working-memory availability and processing depth (Marano et al., 2025). Across models and methods, drafting emerges not as a neutral output stage but as a cognitively diagnostic locus where lexical choice, syntactic structuring, and resource management jointly enact authorship (Hayes, 1996; Kellogg, 2001).

2.2.4 Drafting in the LLM-assisted writing process

LLM-assisted writing redefines drafting from an internally generative process of lexical and syntactic formulation to a distributed activity centered on selection, coordination, and evaluation of externally generated language (Guo et al., 2025; Liu et al., 2024). Rather than constructing sentences incrementally, writers often shift linguistic encoding and idea expansion to the LLM, gaining fluency, speed, and surface-level coherence while reducing engagement in formulation-level decision-making (Zhang et al., 2026; Zhao, 2025). Process-oriented evidence indicates that drafting becomes a supervisory task in which writers prompt the system, evaluate outputs, and decide whether to accept, modify, or discard generated text, shifting cognitive effort from generation to orchestration (Liu et al., 2024; Yao et al., 2025). Some studies frame this redistribution as productive distributed cognition, with LLMs functioning as scaffolds that extend working memory and reduce intrinsic load (Guo et al., 2025; Frenkenberg & Hochman, 2025). Others emphasize cognitive costs, particularly the weakening of memory traces for lexical, syntactic, and structural decisions that traditionally ground authorship (Draxler et al., 2024; Skulmowski, 2024). Consistent with the latter explanation, writers often struggle to explain specific phrasings or structures, reflecting attenuated source monitoring (i.e., reduced ability to distinguish the origins of specific textual elements) and blurred boundaries between human and LLM contributions (Draxler et al., 2024; Qu, 2025). These effects are not inevitable: when drafting is scaffolded to require critical interpretation, selective integration, and explicit justification, metacognitive engagement can be maintained and writer agency preserved despite reduced formulation effort (Lee, 2024; Luo et al., 2025). However, because writing normally involves effortful lexical retrieval and evaluation that encode durable representations of word choice (Hayes, 1996; Kellogg et al., 2013; Li, 2023), offloading these processes to the LLM reduces involvement load and weakens encoding. This results in shallower consolidation of lexical and structural decisions, consistent with accounts linking reduced processing depth to weaker memory traces (Craik & Lockhart, 1972) and with evidence that LLM-assisted writing limits encoding of compositional decisions (Draxler et al., 2024; Liu et al., 2024; Skulmowski, 2024). Consequently, writers may recognize the text as familiar but lack the deep lexical traces required to restore it. Therefore, LLM-assisted drafting remains behaviorally active but is cognitively reconfigured, privileging coordination over generation and reshaping the cognitive foundations of written text production.

2.2.5 Revision in the independent writing process

Revision is a deliberate, resource-intensive process in which writers evaluate text against prior intentions, rhetorical goals, and evolving representations as opposed to merely correcting surface errors (Hayes & Flower, 1980; Kellogg, 1996). Early models conceptualized revision as a recursive reviewing process occurring during or after drafting and requiring access to intended meaning and memory for prior decisions (Hayes & Flower, 1980). Later work reframed revision as a set of goal-driven, task-dependent operations constrained by working memory, motivation, and executive control, and therefore often reduced under high cognitive load (Hayes, 1996; Kellogg et al., 2013; Li, 2023). Across working-memory models, revision is consistently characterized as cognitively demanding, requiring coordination of verbal and spatial representations, retrieval of discourse-level goals, and inhibition of competing alternatives, making it sensitive to individual differences in capacity and expertise (Kellogg, 2001, 2007). Empirical evidence from keystroke logging and stimulated recall shows that revision frequency alone does not predict quality: skilled writers produce fewer but more meaning-oriented and strategically timed revisions, whereas less skilled writers engage in frequent, local, and reactive edits indicative of attentional fragmentation (Révész et al., 2019; Vandermeulen et al., 2024; Xu, 2018). This pattern aligns with writing research showing that global, discourse-level revision depends on retrieval of intended meaning and rhetorical structure, while surface-level revision reflects limited access to prior compositional decisions (Adams et al., 2010; Li, 2023; Sasaki, 2000). Revision is therefore increasingly conceptualized as schema-based problem solving, by which expert writers draw on stored schemas and long-term working memory to diagnose problems, reconstruct goals, and generate alternatives, making revision both metacognitive and reportable (Hayes, 2012). Accordingly, revision is not post hoc cleanup but a cognitively diagnostic activity requiring sustained access to the planning and drafting processes, which are central to writing expertise and authorship.

2.2.6 Revision in the LLM-assisted writing process

Research on LLM-assisted writing shows that revision is cognitively reconfigured toward evaluative coordination of LLM-generated text (Guo et al., 2025; Liu et al., 2024). Across studies, writers frequently engage in iterative cycles of accepting, rejecting, polishing, or reorganizing LLM outputs, yet these revision behaviors are often biased toward surface-level concerns such as fluency, correctness, and stylistic alignment rather than global rhetorical restructuring (Luo et al., 2025; Zhao, 2025). Process-oriented and qualitative evidence suggests that, because the underlying linguistic and structural decisions are externally generated, revision with LLM assistance often does not require retrieval of prior compositional intentions, weakening the memory-based grounding that traditionally supports deep revision and authorship awareness (Draxler et al., 2024; Liu et al., 2024; Skulmowski, 2024). Several studies further indicate that LLM feedback reduces cognitive load during revision, which may increase efficiency and learner confidence but simultaneously reduce opportunities for the reflection necessary to monitor coherence, argumentation, and meaning development (Luo et al., 2025; Zhang et al., 2026). At the same time, not all revision in LLM-mediated contexts is cognitively shallow: when revision is scaffolded through pedagogical guidance that requires writers to articulate goals, evaluate alignment with intended meaning, and justify acceptance or rejection of LLM suggestions, revision can remain metacognitively engaged and meaning-oriented (Lee, 2024; Yao et al., 2025). Nevertheless, without such scaffolding, revision tends to become fragmented and opaque, with writers reporting difficulty tracking idea origins and decision ownership, reinforcing a broader pattern in which revision behavior persists while its cognitive and mnemonic foundations are attenuated (Draxler et al., 2024; Guo et al., 2025). Hence, LLM-assisted revision is behaviorally active but cognitively conditional, supporting surface optimization by default, while requiring carefully designed pedagogical tasks to sustain deeper evaluative engagement and authorship-linked decision making.

2.3 Assessment and the writing process

Writing assessment should be grounded in observable evidence that warrants inferences about underlying cognitive engagement (Deane et al., 2008; Deane, 2011; Fulcher, 2024). Evidence-centered design frames assessment as an explicit evidentiary argument, in which tasks are designed to elicit performances that provide interpretable evidence for claims about unobservable processes such as decision-making, planning, and self-monitoring (Mislevy et al., 2003, 2006). Within this framework, validity depends not on predictive accuracy alone but also on the explanatory adequacy and transparency of the link between observed behavior and the construct it is intended to represent (Chapelle, 2021; Kane, 2013; McNamara, 2001; Messick, 1989). Writing assessment scholars have long cautioned that evaluations based solely on textual artifacts risk underrepresenting the cognitive processes that constitute genuine writing ability (Deane et al., 2008; Deane, 2011; Fulcher, 2024). This concern is amplified in contexts where automated classification systems produce probabilistic judgments that obscure the evidentiary basis for inference (Fraser et al., 2025; Wu et al., 2025). Accordingly, assessments that elicit direct evidence of writers' access to their own planning, drafting, and revision decisions align with construct-valid approaches to writing assessment that emphasize interpretable links between observed performance and underlying cognitive processes, rather than inferences based solely on the final text (Deane et al., 2008; Deane, 2011; Fulcher, 2024).

2.4 Research Gap and Rationale for the Present Study

The reviewed literature reveals a critical gap at the intersection of automated authorship detection, cognitive theories of writing, and construct-valid assessment. Although automated approaches have advanced rapidly, they remain probabilistic, artifact-centered, and vulnerable to failure of robustness in realistic educational contexts, particularly for mixed or assisted authorship (Fraser et al., 2025; Wu et al., 2025). At the same time, generative systems alter writers' experience of authorship, ownership, and self-monitoring, weaken the link between textual production and cognitive engagement, and render declarative authorship claims unreliable (Draxler et al., 2024; Skulmowski, 2024). This shift reflects a redistribution of cognitive effort toward prompt formulation, coordination, and evaluation, while reducing writers' ability to monitor and distinguish their own compositional decisions from those generated by the system (Draxler et al., 2024; Skulmowski, 2024). Although recent work documents how LLMs reshape drafting, revision, and attribution practices (Li & Wu, 2025; Liu et al., 2024; Yao, 2024), relatively few researchers have operationalized writers' cognitive access to lexical, structural, and rhetorical decisions as observable evidence for assessment. In contrast, writing research demonstrates that planning, drafting, and revision in non-LLM-mediated writing are cognitively demanding processes that generate retrievable traces of decision-making, which writers can often articulate and evaluate (Hayes, 1996; Kellogg et al., 2013; Révész et al., 2019; Van Waes et al., 2014). Writing assessment theory similarly emphasizes that valid evaluation requires evidence of the cognitive processes engaged during production (Deane et al., 2008; Deane, 2011; Fulcher, 2024), and that defensible claims must be grounded in observable evidence supporting inferences about underlying cognition rather than textual artifacts alone (Chapelle, 2021; Kane, 2013; McNamara, 2001). However, existing detection systems largely do not engage with this evidentiary logic, relying on classification of text without directly probing access to compositional decisions (Fraser et al., 2025; Wu et al., 2025).

The present study addresses this gap by reframing authorship verification as an assessment problem grounded in cognitive access to the writing process. Rather than inferring authorship from textual features, this study introduces the Content Restoration Authorship Familiarity Test (CRAFT) as a task-based approach to elicit observable evidence of writers' access to prior planning, drafting, and revision decisions. By testing the ability to recall and restore elements of text structure and word choice, the study operationalizes authorship as a cognitive relationship between writer and text, offering a process-oriented alternative aligned with construct-valid principles of writing assessment.

RQ1. To what extent does performance on the Content Restoration Authorship Familiarity Test (CRAFT) discriminate between human-authored and LLM-generated conditions?

RQ2. To what extent do the structural and lexical restoration components of the CRAFT differentially contribute to authorship discrimination across human-authored and LLM-generated conditions?

3. Research Design

This study employed a quantitative, within-subjects experimental design (Creswell & Creswell, 2023) to evaluate the cognitive distinctiveness of human-authored versus LLM-generated text. Following the principles of causal inference outlined by Shadish, Cook, and Campbell (2002), a within-subjects structure was selected to maximize internal validity by controlling for participant-level heterogeneity. By observing the same participant under two distinct production conditions, (1) self-regulated text drafting and (2) LLM-mediated text generation, the design isolates the method of text production as the primary independent variable, effectively neutralizing confounding factors such as L2 proficiency, working memory capacity, and prior topic knowledge.

3.1 Participants

The study was conducted at a large private university in Seoul, South Korea. The participants were 60 undergraduate students majoring in English Education. As pre-service English teachers, the participants possessed high English proficiency (CEFR B2-C1), characterized by advanced metalinguistic awareness and extensive experience with academic writing tasks. All participants were treated as a single cohort for the purpose of the within-subjects comparison, serving as their own controls across the authorship conditions.

3.2 Measure: Content Restoration Authorship Familiarity Test (CRAFT)

The Content Restoration Authorship Familiarity Test (CRAFT; https://github.com/cburst/CRAFT-Content-Restoration-Authorship-Familiarity-Test) was designed to elicit observable evidence of writers' cognitive access to prior text production decisions by requiring participants to restore manipulated elements of their own text. The test operationalizes authorship as the ability to recall and reconstruct lexical and structural choices under conditions where surface familiarity alone is insufficient for successful performance. Each CRAFT instance consisted of two complementary components: (1) structural restoration through sentence-level intruder detection, and (2) lexical restoration through synonym detection and restoration. These components target distinct but related dimensions of writing cognition, with discourse-level organization reflecting planning and revision processes and word-level encoding reflecting drafting and lexical retrieval. The chosen components are consistent with cognitive models that conceptualize writing as a recursive, working-memory-constrained interaction of planning, formulation, and revision, focusing on decisions that are encoded and remain potentially retrievable (Hayes, 1996; Kellogg et al., 2013).

3.2.1 Structural restoration: sentence intruders

To assess participants' access to discourse-level structure, each participant's text was individually modified to include typically four inserted intruder sentences, with the exact number adjusted downward in cases where shorter source texts did not permit four plausible insertions.

These intruder sentences were LLM-generated to be topically plausible and locally coherent. Intruder sentences were variably distributed throughout the text to avoid positional predictability, and were designed to resist elimination through superficial cues such as lexical mismatch or grammatical anomaly. Participants were informed that the typical number of intruder sentences was four, but that some texts might contain fewer, and were instructed to identify all sentences not found in the original text rather than selecting a fixed number. Because this task permits guessing, incorrect selections were penalized (−1 point per incorrect intruder selection), while correct identifications were awarded positive points. This asymmetric scoring structure was implemented to discourage over-selection strategies and to ensure that performance reflects genuine structural familiarity rather than strategic guessing. This component targets the writer's ability to recognize deviations from their own discourse and organizational decisions, which are assumed to be encoded during composition, and retrievable during post hoc evaluation (see Adams et al., 2010; Galbraith & Torrance, 2004; Révész et al., 2019; Van Waes et al., 2014).

3.2.2 Lexical restoration: synonym reconstruction

To assess participants' access to lexical-level decisions, each text included 10 target lexical items, comprising five replacement-original word pairs. For each pair, the original lexical item produced by the writer was replaced within the text by a contextually plausible synonym (e.g., confidently/assertively), such that the replacement appeared in the sentence and the original word was removed. Participants were required to identify both (a) which in-text words were replacements and (b) the corresponding original lexical items. To constrain the response space while preserving task difficulty, participants were provided with the first letter of each original word (only the replacement appears in the text) and the part of speech for each replacement-original pair. All synonym candidate words were selected to be semantically appropriate within the sentence context and matched for part of speech, morphological form, grammatical function, and general frequency band (operationalized as frequency counts in the Wikipedia corpus), thereby minimizing the possibility of correct responses based on salience, rarity, or grammatical mismatch. Additional constraints were applied to avoid transparent substitutions (e.g., highly marked or obscure vocabulary) and to ensure that alternatives could be easily confused at the surface level. Unlike the sentence intruder detection task, the synonym restoration component was intentionally designed to be somewhat difficult and to rely on fine-grained lexical memory. As such, no penalty was assigned for incorrect responses, and scoring was based solely on correct (or nearly morphologically correct) identifications. This approach was adopted to prevent excessive penalties for partial knowledge, and so that performance would reflect graded access to lexical encoding rather than risk-averse responses. This design reflects models of writing that treat lexical selection during drafting as a resource-intensive process that produces durable, retrievable representations of word choice under working-memory constraints, making successful restoration an indicator of prior involvement in lexical selection (see Kellogg et al., 2013; Li, 2023).

3.2.3 Design rationale

The combined use of structural and lexical manipulations reflects the multidimensional nature of writing cognition, with sentence intruders targeting discourse-level organization associated with planning and revision, and synonym restoration targeting lexical encoding processes associated with drafting (see Hayes, 1996; Kellogg et al., 2013). Task constraints were designed to preclude solutions via surface-level heuristics, requiring access to internal representations formed during composition, consistent with models of writing in which planning, drafting, and revision produce durable, retrievable representations of compositional decisions (see Hayes, 1996; Kellogg et al., 2013). The asymmetry in scoring between components reflects differences in task characteristics. Sentence intruder detection involves a closed set with higher susceptibility to guessing, justifying the use of penalties to control for guessing that does not reflect underlying knowledge; whereas synonym restoration involves open-ended or high-uncertainty discrimination, where incorrect responses are more likely to reflect task difficulty than strategic guessing. These design features position CRAFT as a process-oriented assessment instrument that elicits interpretable evidence relevant to authorship by directly probing writers' access to prior compositional decisions, consistent with evidence-centered approaches to assessment (Deane et al., 2008; Fulcher, 2024; Mislevy et al., 2006).

3.3 Procedure

The study was conducted over two consecutive weekly sessions aligned with the course writing schedule. In the first session, participants completed a handwritten composition task in response to a weekly writing prompt. Participants were instructed to produce approximately 16 sentences on paper. To ensure that the resulting text reflected independent writing processes, all participants' digital devices were removed during completion of the task. Immediately following composition, participants photographed their handwritten texts and produced a typed transcription. The CRAFT Python script was applied to each transcribed text (which took approximately one minute per student) to generate individualized test forms containing structural and lexical manipulations. These tests were printed and administered in the same session. Participants were given 10 minutes to complete the CRAFT tests for their own handwritten composition.

In the second session, conducted one week later, participants completed a parallel CRAFT task based on an LLM-generated text. The topic of each participant's original handwritten composition was extracted and used to generate a new 16-sentence text, to which the CRAFT script was applied to produce individualized test forms. Participants were again given 10 minutes to complete the tests. The one-week delay was intentional, reflecting the absence of prior encoding rather than decay of memory traces, as participants had not engaged in the original compositional process. This two-session design enabled a within-subject comparison between performance on a CRAFT test derived from a participant's own handwritten text and a structurally parallel test derived from an LLM-generated text on the same topic. By holding topic constant and varying only production mode, the design isolates cognitive engagement in authorship, allowing differences in performance to be interpreted as differential access to underlying compositional decisions.

3.4 Data Analysis

3.4.1 Scoring Protocols

For the structural restoration component (sentence intruder detection), scoring was binary. Participants received one point for each correctly identified replacement sentence. Incorrect selections (i.e., mistakenly identifying an original sentence as a replacement sentence) were penalized (−1 point) to discourage over-selection and guessing strategies. For the lexical (synonym) restoration component, responses were scored using a strict but morphologically tolerant rubric. A response was marked correct if the participant successfully retrieved the original lexical item. To account for minor retrieval variation not indicative of memory failure, full credit was also granted for morphologically related forms (e.g., 'analysis' for 'analyze'), provided the semantic root was preserved. Total CRAFT scores were computed by summing performance across both components. For the structural restoration component (intruder identification), scores ranged from −4 to +4, reflecting penalties for incorrect selections and rewards for correct identifications, while the lexical restoration component scores ranged from 0 to 10, reflecting the number of correctly restored lexical items.

3.4.2 Analytical Strategy

Given the within-subject design, analysis focused on paired comparisons of participant performance across the two text conditions (self-authored vs. LLM-generated). Descriptive statistics (mean, median, range, and standard deviation) were computed separately for each condition and for each component (sentence intruder detection and synonym restoration), as well as for the total CRAFT score. To evaluate differences between conditions, paired-samples t-tests were conducted on component scores and total scores. Effect sizes were reported as Cohen's d for paired samples. In addition, paired difference scores were examined descriptively to characterize the magnitude and consistency of the observed condition effect.

Analyses were conducted at two levels, corresponding to the research questions. First, total CRAFT scores were analyzed to evaluate overall discriminative capacity between conditions (RQ1). Second, component-level performance was examined to assess the relative contribution of structural and synonym restoration to between-condition differences (RQ2). This approach enables examination of both overall condition effects and the differential contribution of component processes to authorship discrimination.

4. Results

4.1 Overall CRAFT Performance Across Authorship Conditions (RQ1)

In addressing RQ1, CRAFT performance differed substantially between self-authored and LLM-generated conditions, as seen in Table 1 and Figure 1, which show clear separation with minimal overlap between the distributions. A clear separation in total scores was observed ($\Delta = 12.78$, $d = 5.34$), indicating a large and consistent advantage for self-authored texts, with narrow confidence intervals reflecting stable differences across participants. This pattern supports the strong discriminative capacity of the CRAFT for distinguishing between human-authored and LLM-generated texts.

Table 1

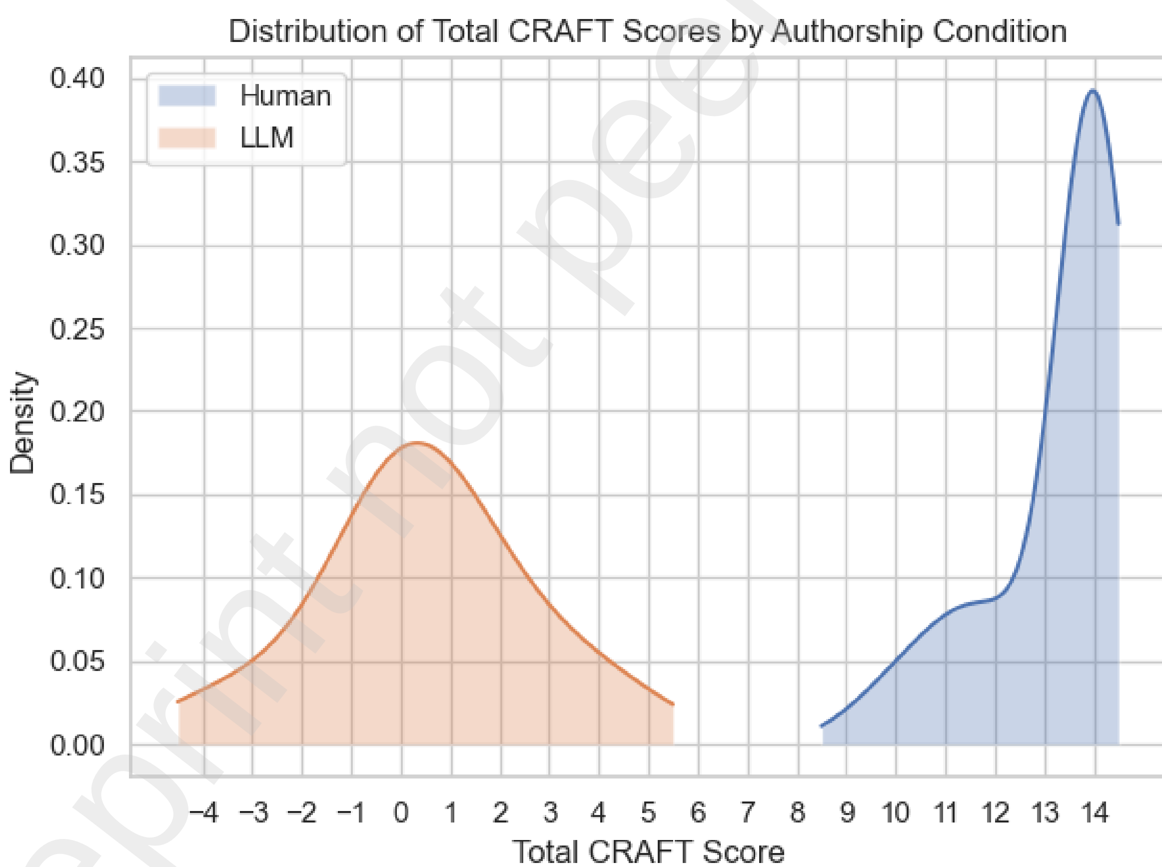
CRAFT Performance by Condition

Measure	Human M (SD)	LLM-generated M (SD)	Δ M	Cohen's d	95% CI	N
Structural Restoration	3.97 (0.26)	-0.32 (1.32)	4.28	3.16	[3.93, 4.63]	60
Lexical Restoration	9.28 (1.33)	0.78 (1.30)	8.50	4.65	[8.03, 8.97]	60
Total	13.25 (1.36)	0.47 (2.09)	12.78	5.34	[12.16, 13.40]	60

Note. Values are means with standard deviations in parentheses. Δ M = mean paired difference (Human - LLM-generated). Cohen's d reflects paired effect size. CI = confidence interval.

Figure 1

Distribution of Total CRAFT Scores by Authorship Condition



Note. Density curves represent smoothed distributions of scores.

4.2 Differential Contributions of CRAFT Components Across Authorship Conditions (RQ2)

To address RQ2, component-level performance indicated that both structural and lexical restoration discriminated between conditions, with higher scores in the self-authored condition, as seen in Table 1. Synonym restoration showed larger differences ($\Delta = 8.50$, $d = 4.65$) than sentence intruder detection ($\Delta = 4.28$, $d = 3.16$), indicating a stronger lexical signal despite differing score ranges. Although each component showed partial overlap, their combination produced complete separation, suggesting that complementary structural and lexical evidence is necessary for robust authorship discrimination.

5. Discussion

The present study examined whether human authorship can be identified through process-based evidence of engagement with a text rather than probabilistic classification of textual features. Across analyses, CRAFT performance differed substantially between self-authored and LLM-generated conditions, with very large effect sizes and complete separation in total scores. These findings support the view that authorship is not a property of the final text alone, but a cognitive relationship between writer and text (Hayes, 1996; Kellogg et al., 2013) that can be operationalized through access to prior compositional decisions. In an evidence-centered design framework (Mislevy et al., 2006), CRAFT performance can be interpreted as evidence of human authorship insofar as successful recall and restoration depend on access to internal representations formed during planning, drafting, and revision, rather than primarily on surface familiarity (Hayes, 1996; Kellogg et al., 2013; Révész et al., 2019; Van Waes et al., 2014).

With respect to RQ1, the CRAFT reliably distinguished between human-authored and LLM-generated texts when administered immediately after production. This directly addresses limitations of automated detection, which yields probabilistic, context-sensitive judgments with limited robustness, interpretability, and resistance to adversarial adaptation in educational contexts (Fraser et al., 2025; Kehkashan et al., 2025; Pudasaini et al., 2025; Wu et al., 2025; Yang et al., 2025). Rather than inferring authorship from surface-level patterns, the CRAFT elicits observable evidence of access to planning, drafting, and revision decisions, consistent with construct-valid approaches to assessment grounded in cognitive processes (Deane et al., 2008; Deane, 2011; Fulcher, 2024; Mislevy et al., 2006). The substantial separation in total scores for human and LLM texts indicates that performance differences reflect differential access to encoded compositional decisions rather than variation in surface textual quality (Hayes, 1996; Kellogg et al., 2013).

With respect to RQ2, structural and lexical components contributed differentially to authorship discrimination. Synonym restoration produced larger effects, consistent with models treating lexical selection as a resource-intensive process that generates durable, retrievable traces (Kellogg et al., 2013; Li, 2023). Sentence intruder detection, while still highly discriminative, produced smaller effects and partial overlap, suggesting that discourse-level structure, though encoded during planning (Hayes & Flower, 1980; Kellogg, 1996), may be more amenable to restoration through coherence expectations or schema-based reasoning (see Hayes, 2012; Révész et al., 2019). The combination of both discrimination methods yielded substantial separation, indicating that authorship is best captured through multiple complementary traces of the writing process (Hayes, 1996; Kellogg et al., 2013).

These findings align with research on LLM-assisted writing showing that observable writing behavior can persist despite reconfigured underlying cognitive processes (Draxler et al., 2024; Skulmowski, 2024). Reduced performance in the LLM condition is consistent with evidence that LLM-assisted drafting reduces engagement in lexical retrieval and evaluation, weakening memory traces associated with prior lexical and structural decisions (Liu et al., 2024; Zhang et al., 2026). Accordingly, the CRAFT functions as a direct probe of cognitively encoded writing decisions rather than a detector of textual artifacts. More broadly, the results support a shift from viewing authorship as a textual attribute to treating it as an observable consequence of cognitive engagement during writing, consistent with evidence-centered assessment frameworks emphasizing interpretable links between performance and underlying constructs (Chapelle, 2021; Kane, 2013; McNamara, 2001; Messick, 1989).

6. Limitations

First, effort effects may have influenced performance in the self-authored condition. Planning depth, revision behavior, and lexical encoding are sensitive to cognitive load, motivation, and task engagement (Deane et al., 2008; Hayes, 1996; Kellogg, 1996). Because compositions were completed in a low-stakes classroom context, some participants may not have fully engaged in planning and drafting processes necessary to generate robust, retrievable representations of compositional decisions, which depend on sustained cognitive engagement and working-memory resources (Hayes, 1996; Kellogg et al., 2013). Reduced engagement would weaken encoding and potentially lower CRAFT performance even for genuinely self-authored texts, suggesting that observed differences may underestimate the effect under conditions of full engagement.

Second, the CRAFT remains susceptible to lexical salience effects, whereby highly salient or domain-specific items may be recognized independent of the writing process. This concern aligns with research on lexical accessibility and frequency, where retrieval may reflect prior exposure rather than encoding during composition (Li, 2023). Although constraints were implemented to minimize such effects, some items may have remained easier to guess than intended. While a revised test-generation script incorporates additional filtering procedures, further refinement is needed to align item selection with controlled lexical encoding. Finally, the study's relatively homogeneous sample of advanced L2 university students limits generalizability. Given that writing processes vary across proficiency levels and contexts (Barkaoui, 2019; Li, 2023; Révész et al., 2019), further clarification is needed as to whether similar separation would be observed in shorter texts, different proficiency groups, different patterns of L2 encoding, or different writing environments.

7. Directions for Future Research

The present study should be understood as a proof of concept, demonstrating the feasibility of process-based authorship verification under controlled conditions and suggesting several directions for future research. First, further work is needed to refine task design through expanded piloting, particularly across diverse learner populations and proficiency levels. Given that writing processes and working memory constraints differ substantially across contexts (Kellogg et al., 2013; Li, 2023), systematic variation in task parameters will be necessary to establish generalizable design principles. Second, longitudinal research is needed to examine the stability of individual baselines, particularly in relation to changes in writing proficiency and familiarity with LLM tools. Given that writing expertise involves the development of more efficient and structured retrieval processes (Kellogg, 2001; Hayes, 2012), CRAFT performance may evolve over time in ways that reflect broader cognitive development. Finally, future work should explore scalability and integration, including digital administration, automated item generation, and alignment with classroom assessment practices. Such developments would be necessary to position process-based authorship verification as a practical alternative to existing detection systems.

A related direction is the conceptualization of authorship as a continuum rather than a binary condition. The present design establishes two anchor points, fully self-authored and fully LLM-generated texts, which may be interpreted as endpoints of a broader authorship spectrum. Future research should investigate hybrid authorship conditions in which texts are produced with varying degrees of LLM assistance, and test whether CRAFT performance varies proportionally across these conditions. Prior research suggests that LLM use redistributes cognitive effort toward coordination and evaluation rather than internal generation (Draxler et al., 2024; Liu et al., 2024), which should result in intermediate levels of lexical and structural recall for partially LLM-assisted texts. In this framework, the performance gap between self-authored and LLM-generated conditions can be treated as a bounded range including degrees of text hybridity. Empirically establishing whether CRAFT scores scale systematically across this range would allow for the identification of interpretable regions of authorship, where scores closer to the self-authored endpoint may reflect hybrid approaches that preserve cognitively meaningful engagement with planning, drafting, and revision, while scores closer to the LLM-generated endpoint may indicate increasing reliance on externally generated text with reduced access to underlying compositional decisions. Such distinctions would move authorship verification beyond binary classification toward graded, process-based interpretations with direct implications for assessment validity and instructional policy.

8. Implications for Practice

The findings suggest that process-based authorship verification can be operationalized in ways that are both interpretable and pedagogically meaningful, and conducted in an automated manner. Rather than relying on opaque detection systems that produce probabilistic judgments (Fraser et al., 2025; Wu et al., 2025), instructors can use tools such as the CRAFT to directly assess students' engagement with their own writing processes. One key application is the establishment of individual baseline profiles, consistent with evidence-centered approaches to assessment that emphasize individualized inference based on observable performance (Chapelle, 2021; Mislevy et al., 2006). By administering the CRAFT under both self-authored and LLM-generated conditions, instructors can derive reference points that reflect each student's typical level of cognitive engagement. These baselines can then be used to interpret future work, allowing instructors to estimate the extent to which a given text reflects internally generated versus externally mediated composition. Importantly, this approach reframes authorship verification as a matter of evidence-based evaluation rather than surveillance, aligning with pedagogical models that emphasize planning, drafting, and revision as core components of writing development (Deane et al., 2008; Hayes, 1996; Kellogg et al., 2013). In doing so, CRAFT offers a more transparent and educationally constructive alternative to enforcement-based approaches used to preserve academic integrity.

In practice, such baseline profiles may also serve as a foundation for formative dialogue rather than punitive enforcement. When discrepancies arise between a student's prior CRAFT performance and subsequent written work, instructors can use this evidence to initiate structured discussions about writing processes, decision-making, and tool use. Rather than positioning authorship verification as a binary judgment, this approach supports reflective engagement by prompting students to explain how a text was produced and to articulate their level of involvement in its development. This approach may also mitigate the adversarial dynamics often associated with authorship disputes. Because the evidentiary basis is grounded in each student's own prior performance, the burden shifts from instructor-driven suspicion to student-centered explanation, reducing reliance on opaque detection tools and external judgments. At the same time, the emotional impact of such assessments must be carefully considered. Transparent communication, low-stakes implementation, and opportunities for revision are essential to ensure that process-based verification supports learning and accountability without undermining student trust or motivation. This approach also aligns with broader principles of assessment for learning, which emphasize the role of assessment in supporting long-term development rather than merely evaluating performance (Boud & Falchikov, 2006).

9. Conclusion

The present study examined whether authorship can be distinguished through process-based evidence of engagement with a text rather than probabilistic classification of textual features. Across analyses, CRAFT performance differed substantially between self-authored and LLM-generated conditions, with very large effect sizes and substantial separation in total scores. These findings indicate that differential access to encoded compositional decisions—operationalized through the ability to recall and restore word choice and text structure—provides a robust and interpretable basis for distinguishing modes of text production.

More broadly, the study supports a reconceptualization of authorship as a cognitive relationship between writer and text, grounded in planning, drafting, and revision (Hayes, 1996; Kellogg et al., 2013), rather than a property of textual output. In contrast to probabilistic detection approaches based on surface features (Fraser et al., 2025; Wu et al., 2025), process-oriented methods such as the CRAFT align with construct-valid assessment by eliciting direct evidence of cognitive engagement (Deane et al., 2008; Fulcher, 2024). The findings further suggest that authorship may be better understood as a continuum, reflected in varying levels of access to prior compositional decisions. Although further research is needed to refine task design and extend findings across contexts, the results demonstrate that process-based authorship verification is both feasible and theoretically grounded, offering a principled and educationally meaningful alternative to artifact-based detection in LLM-assisted writing contexts.

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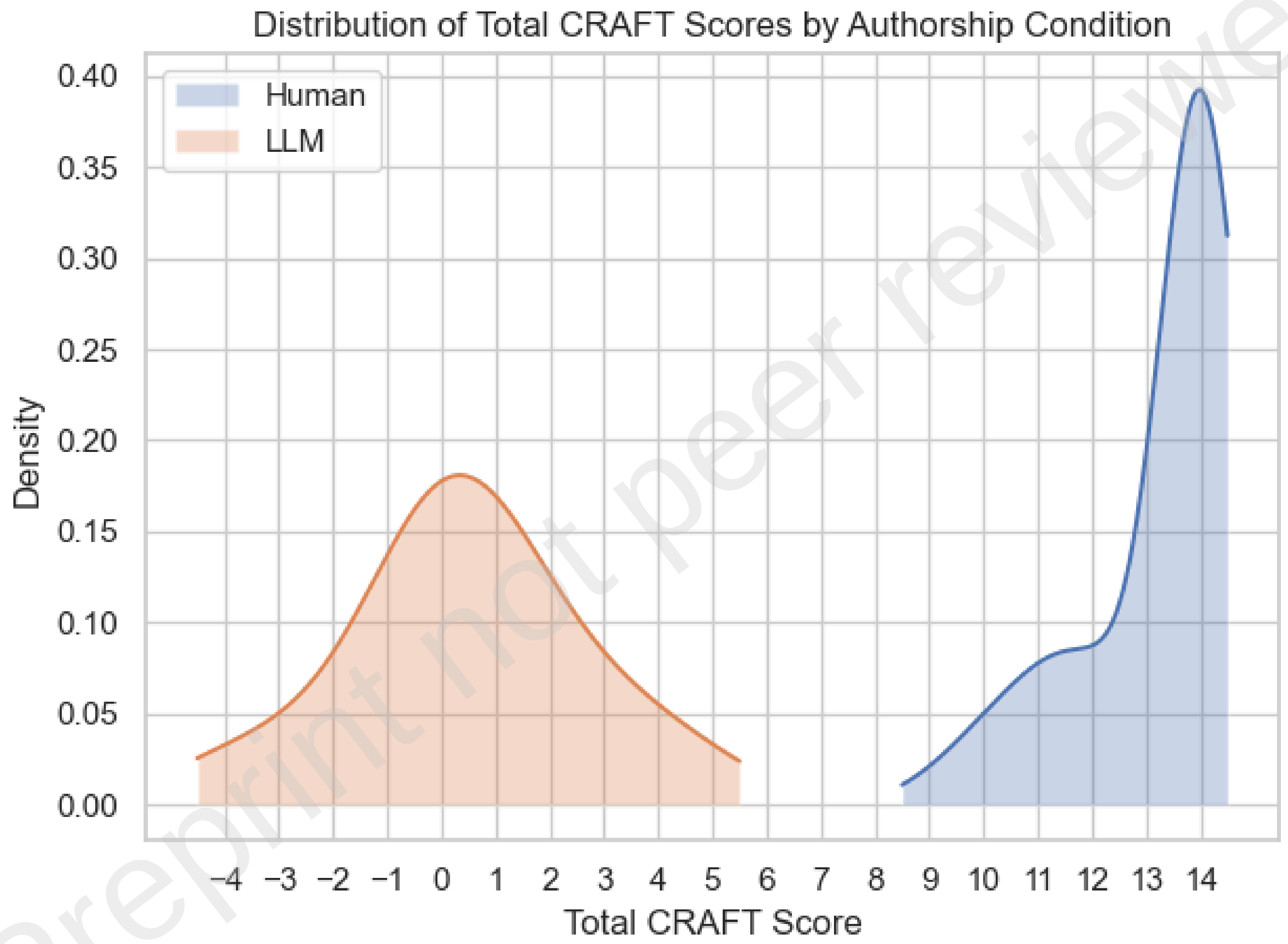
Distribution of Total CRAFT Scores by Authorship Condition
Human
LLM
Density
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4 −3 −2 −1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Total CRAFT Score